

Aim

an arduino uno based project that uses an ir sensor to detect an object or person and automatically opens and close gate or door using a servo motor

Components required

- arduino uno
- ir sensor module
- sg90 servo motor
- breadboard
- jumpwire
- usb cable

Connections

- The arduino is used as the main control unit in the circuit .
- The IR sensor module is connected to the arduino with VCC to 5 V , GND to GND and out to digital pin D8
- The SG90 servo motor is connected to the arduino to perform the required mechanical movement .
- All components are assembled on a bread board using jumper wire and the arduino is powered / programmed through a USB cable

Working

- The IR sensor continuously emits infrared light and monitors for reflection
- When an object comes near , the reflected IR is detected and the sensor's OUT pin signal changes state .
- The arduino reads this digital signal.
- Tf an object is detected , the arduino sends a PWM signal to the servo motor , rotating it to the open position .
- After a short delay , the servo returns to its closed position .

